

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	HAZOP Study Report
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HAZOP Study Report

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Revision History

Rev No.	Date	Description of Revision
R01	14-07-22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-004

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
ALARP	As Low As Reasonably Practicable
DEP	Design and Engineering Practice
F & G	Fire & Gas
HAC	Hazardous Area Classification
HAZOP	Hazard and Operability Studies
PEFS	Process Engineering Flow scheme
PFS	Process Flow scheme
SOP	Standard Operating Procedure

1. Introduction

1.1 Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

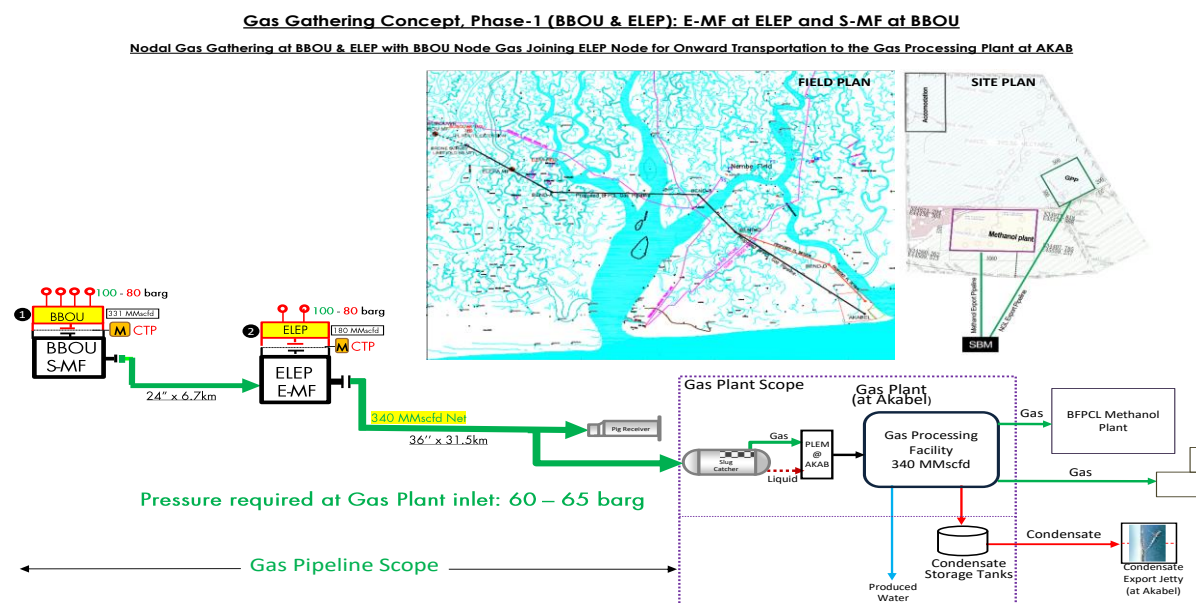


Figure 1: Gas Gathering Concept, Phase 1

1.2 Purpose

The purpose/objectives of the HAZOP study are to:

- Assess risks and control measures in the design, to ensure risks are adequately managed to ALARP
- Recommend suitable actions towards risk reduction if the controls and recovery measures are not deemed to be sufficient to control the hazard
- Identify operability and maintainability issues in the design
- Prepare a report of the findings of the study

The FEED complied with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 80.00.00.15-Gen	Hazard and Operability (HAZOP) Study

The latest versions of the regulations, codes, and standards and extant amendments or supplements was applied. Local laws regulating the oil and gas industry in Nigeria was strictly adhered to. The order of precedence for the documents was as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent was apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof was referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Methodology

A HAZOP is a structured analysis of a process, system or operation carried out by a multi-discipline team of competent people and chaired by an independent person. The team carried out a systematic examination of the processes and operations of the facilities covered within the project scope. This was aided by using a set of guidewords in combination with parameters to give meaningful deviations from the design intent. The team concentrated on those deviations that could lead to:

- Potential hazards to safety, health, or the environment; and
- Operability problems.

Where realistic causes of deviation was found, the team evaluated the consequences based on the team's experience and judgment. Where the existing safeguards are deemed inadequate, the team recorded an action for change or recommended further investigation of the problem.

3.1 Guidewords

HAZOP guidewords (Deviation + Parameter) were adopted for the review session. Typical guidewords are shown as Table 1. The guidewords were intended to prompt discussion and not intended to be exhaustive

Table 1: HAZOP Guideword

GUIDEWORD	PROPERTY	POSSIBLE CAUSES
NO	Flow	Wrong routing, blockage (debris, hydrate, sand), slip blind left in, faulty non-return valve, burst pipe, control valve, isolation valve, pump or vessel failure.
MORE	Flow	More than 1 pump operating, reduced delivery head, increased suction pressure, lower downstream pressure, other routes, exchanger tube leaks, greater density, gas break through, valve malfunction.
	Pressure	Surge, relief, leakage from an HP connection (lines and flanges), thermal, rate of pressurising lines, blockage downstream.
	Level	Surge, blocked outlet, control valve failure (see also More Flow)
	Temperature	Fouled cooler, cooling water failure, failed exchanger tubes, external fire.
	Viscosity	Change of material, specification or temperature.
LESS	Flow	Partial blockage, vessel or valves failing, leaks, loss of pump efficiency.
	Pressure	Generation of vacuum by pump or drain out of vessels, condensation from vapour or gas dissolving in a liquid. Blocked pump or compressor suction.
	Level	Leaks, control valve failure, upstream upset, blockage
	Temperature	Freezing, loss of pressure (Joule Thompson effect), loss of heating, failed exchanger tubes.
	Viscosity	Change of material, specification, or pressure.
REVERSE	Flow	As for NO FLOW, plus emergency venting, 2-way flow.
PART OF	Composition change	Passing through isolations, leaking exchanger tubes, wrong feed material. Phase change, out of specification.
AS WELL AS	Contamination	H ₂ S, mercury, radioactive nuclides, wax, sand, oxygen in water.
OTHER THAN	Relief	Relief valve, bursting disc, location of discharge, bellows type, fire.
	Sampling	Type, cooling of, response time.
	Corrosion and Erosion	Material specifications, fluid velocities, baffles, pipe-guides, hangers, anchors, external attack, sulphide/chloride stress corrosion, zinc embrittlement.
	Service Failures	Cooling Water, instrument air, steam, nitrogen or inert gas; power to machines, control circuits and computers.
	Maintenance	Isolation, drainage, purging, cleaning, drying, access, incorrect installation of equipment, hydrotest, lifting.
	Static	Electrical grounding, low conducting liquids, insulated conductors.
	Outside conditions	Fire, weather, environmental change, impact flood, etc.
	Operation	Start up, shut down (normal, emergency), maloperation, pigging, layout, different modes.
	Layout	Ergonomics issues, (raised once per unit)

3.2 Preparation Work, Nodes, and Documentation

The Nodes of study are shown in Table 2 below. The marked-up drawings are shown in Appendix 2

Table 2: HAZOP Nodes

Node	Title	Drawing No.
1	BBOU MF Receiver & Launcher	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_1 A01; EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_2 A01
2	BBOU Open Drain & Closed Systems and Vent System	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_3 A01
3	ELEP MF Receivers & Launcher	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_4 A01; EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_5 A01; EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_6 A01
4	ELEP Open Drain, Closed Drain System and Vent System	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_7 A01
5	AKABEL Pig Receiver	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_8 A01

3.3 Preparation Work, Nodes, and Documentation

The following key documentation was used for the HAZOP study

- Approved for HAZOP Process Flow Diagram(s)
- Approved for HAZOP Process Engineering Flow Scheme (PEFS)
- Approved for HAZOP Process Control Narratives & Safeguarding Memorandum
- Plot plan
- Flowing and Closed In Tubing Head Pressure of the NAG wells at BBOU and ELEPA

Other documents kept by the Design Team for reference purposes

- Process and F & G Cause & Effect Matrices
- Process Safeguarding narratives
- Materials selection report(s)
- Vent dispersion calculation report
- Fire & Explosion Risk Assessment reports
- Hazardous Area Classification (HAC) drawings
- Operations Manual

3.4 HAZOP Agenda

The agenda was as follows:

- Team introduction, domestic arrangements, safety instructions
- HAZOP Chairman (Facilitator) introduced the scope of the HAZOP study
- Design Team Lead Engineer presented the process description of the project, highlighted the key features and design decisions taken
- The HAZOP Chairman explained the HAZOP process including the HAZOP worksheet format and terminologies
- The Chairman:
 - Proposed Nodes and agree on Nodes with the team.
 - Commenced HAZOP using the guidewords. This was not a rigid list and the team frequently added additional hazards not covered by these guidewords as required.
 - All relevant information was recorded on the worksheet.
 - Refreshments & comfort breaks was held at intervals as agreed by the team.

3.5 HAZOP Session Details

The HAZOP was carried out on the 8th of July 2022, commenced at 09.00 AM and finished by 4:00 PM same day, virtually. Attendees/ reps are shown as in Table 2 below.

Table 3: HAZOP Attendee List

Discipline	Name	Company
ESLP Consultants Reps (HAZOP Facilitator)		
HAZOP Chairman	Emeka Maduabuchi	ESLP Consultants Ltd
HAZOP Scribe	Valentine Orafu	ESLP Consultants Ltd
Operations Safety	Ekene Okoye	ESLP Consultant Ltd
NUPRC Reps (Regulator)		
Asst Chief Chemist	Hamu Adamu	NUPRC
Mechanical	Francis Lennin	NUPRC
BRASS FERTILIZER AND PETROCHEMICAL COMPANY LIMITED (Project Owner)		
GM, Upstream/Midstream Integration	Ralph Afolabi	BFPCL

Discipline	Name	Company
TCE (Project Management Consultant for BFPCL)		
Project Management	Sandeep Mithailal Yadav	TCE
Project Management	D.P. Mishra	TCE
EPIC ATLANTIC Reps (FEED Contractor)		
Project Director/Process	Aloy Duru	EPIC ATLANTIC
Project Management	Maduka Ekpunobi	EPIC ATLANTIC
Project Services	Obima Oragwu	EPIC ATLANTIC
Operations Readiness & Assurance	Olarinde Ige	EPIC ATLANTIC
Process Automation, Control and Optimisation	Paul Roy	EPIC ATLANTIC
Process	Obinna Joseph	EPIC ATLANTIC
Pipelines, Mechanical, Piping	Emeka Nwachukwu	EPIC ATLANTIC
Materials & Corrosion	Johnson Lantey	EPIC ATLANTIC
Project Management	Shedrack Anigbankpu	EPIC ATLANTIC
Process	Basil Akpan	EPIC ATLANTIC
Civil/Structural	William Irem	EPIC ATLANTIC
TREWAN Reps (Interface Consultants to BFPCL)		
Interface	Tunji Olukayode	TREWAN
Interface	Ololade Shakioye	TREWAN
SPDC Reps (Gas Supplier)		
Surface Lead	Obioma Iwuchukwu	SPDC
Execution	Othuke Okpokoro	SPDC
Pipelines	Unyimeabasi P. Aghadiuno,	SPDC
IM/IT	Emy Ngoye	SPDC
Metering	Wisnu Purwanto	SPDC

The HAZOP was recorded using PHApro 8.5.1.0 Software. The worksheet was projected onto a screen so that team members have an opportunity to review and comment during the session. Actions recorded during the session was assigned to individual team members for follow-up and closeout after the meeting

3.6 HAZOP Recommendation

In total, 24 recommendations were made – HAZOP worksheet in Appendix 1. The full list of the recommendations is shown Table 3 below. Action parties will be identified by the Project team, who will also set priority setting on the actions, as required.

Table 4: HAZOP Recommendation

S/N	Recommendations	Place(s) Mentioned
1	Confirm the overpressure protection system on SPDC gathering manifold to ensure that it is adequate to protect 900lb class piping	Causes: 3.1.1
2	Carry out ELEPA MF vent dispersion and thermal radiation modelling to determine safe dispersion distances	Causes: 3.2.1
3	Install flame arrestor on the ELEPA MF vent tip	Causes: 3.2.1
4	Provide a check valve on the SPDC BBOU and ELEPA tie-in points into the common manifold (Class 1 check valve)	Causes: 1.4.1
5	Make MV-1001 LO	Causes: 1.1.4
6	Carry out 3D model review for the entire project facility during DED	Causes: 6.1.1
7	Confirm noise levels from BDV-CDT-001 and develop noise map as appropriate during DED	Causes: 1.20.1
8	Consider removing the valves E2R on vent line of BBOU-CDT-001	Causes: 2.1.1, 2.1.2, 4.1.1, 4.1.2
9	Lock open E2R on the inlet to the BBOU-CDT-001	Causes: 2.1.1
10	Lock open E2R on vent line of BBOU-CDT-001	Causes: 2.1.2
11	Provide high level protection on BBOU-CDT-001	Causes: 2.11.1
12	Develop draining procedure for the pig trap	Causes: 2.11.2, 4.11.2
13	Provide a check valve on all tie-in points into ELEPA manifold (class 1 check valve)	Causes: 3.4.1
14	Provide the means for positive isolation of the incoming lines into ELEPA manifold	Causes: 6.15.1
15	Provide a means of manual depressurisation of the ELEPA header	Causes: 6.15.2
16	Lock open E2R on the inlet to the ELEP-CDT-001	Causes: 4.1.1
17	Lock open E2R on vent line of ELEP-CDT-001	Causes: 4.1.2
18	Provide high level protection on ELEP-CDT-001	Causes: 4.11.1
19	Confirm the overpressure protection system on SPDC gathering manifold to ensure that it is adequate to protect the BBOU 900lb class piping	Causes: 1.1.1

20	Carry out a vent dispersion and thermal radiation modelling other vent system to determine safe dispersion and thermal radiation distances	Causes: 1.2.1
21	Carry out hazardous area classification of the entire project facilities	Causes: 6.18.1
22	Develop Emergency Spill Response Procedure/Plan for the project facilities	Causes: 6.18.1
23	Provide fire detection system in all the manifold	Causes: 6.18.1
24	Install flame arrestor on the BBOU vent tip	Causes: 1.2.1

Appendix 1: HAZOP Worksheet

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
1.BBOU MF Receiver & Launcher	1.No Flow	1.Blockage on inlet piping from SPDC BBOU wells (Fail close of SDV S1001/S1002: Inadvertent closure of any of the manual valves MV-1043, MV-1045, MV-1022, MV-1027)	1.Pressure build up upstream the blockage up the CITHP of SPDC BBOU well (190barg) potentially leading to rupture of the piping and LOPC, resulting into injuries/fatalities from ignited release and environmental pollution from unignited release (Piping is 900lb class)	5	C		1.Overpressure protection system from SPDC gathering manifold 2.PSV-1001 set at 110barg (fire case) 3.Pressure trip tied to ESD 4.PM 5.SOP	19 Confirm the overpressure protection system on SPDC gathering manifold to ensure that it is adequate to protect the BBOU 900lb class piping
		2.Blockage on the relief line (inadvertent closure of	1.Loss of relief functionality when demand is placed on	5	B		1.Valves MV-1003/1006 are LO	

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
		manual valve MV-1003, MV-1006)	BDV-B-1001, potentially leading to rupture of the line and LOPC					
		3.Blockage on any of the manual relief valves header	1.Loss of blow down functionality in a fire case scenario, potentially leading to rupture of the line and LOPC	5	B		1.Valves are LO	
		4.Blockage on the chemical injection line (inadvertent closure of MV-1001)	1.Loss of chemical inhibition injection potentially leading to increasing corrosion rate of the piping/pipe line	4	B		1.SOP	5.Make MV-1001 LO
	2.More Flow	1.Fail open of BDV-B-1001	1.More gas to the close drain and vent system potentially leading to GHG emission	1	C		1.PM	20 Carry out a vent dispersion and thermal radiation modelling other vent system to determine safe dis-

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
								person and thermal radiation distances
			2.Potential low temperature on the downstream piping (7 deg C)	1	C		2.Piping is CS, LTD is -29degC	24 Install flame arrester on the BBOU vent tip
		2.Inadvertent opening of MV-1044 on the flow meter bypass	1.Metering error	1	B		1.SOP	
	3.Less flow	1.Passing relief valves (PSV-1001)	1.More gas to the vent system potentially leading to GHG emission	1	C		1.PM	
	4.Reverse Flow	1.Possible reverse flow from a future OPUG-BENE manifold into SPDC BBOU when SPDC BBOU is down	1.Potential process safety issues if there is maintenance work in SPDC BBOU system	5	B		1.	4.Provide a check valve on the SPDC BBOU and ELEPA tie-in point into the common manifold (Class 1 check valve)

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	5.As well as Flow	1.N/A						
	6.Part of Flow	1.N/A						
	7.Other than Flow	1.N/A						
	8.More Pressure	1.See flow Scenarios						
	9.Less Pressure	1.see LOPC on the node						
	10.No Level	1.N/A						
	11.More Level	1.N/A						
	12.Less Level	1.N/A						
	13.More Temperature	1.Pool fire on the platform	1.Potential pressure build up on the piping possibly leading to rupture and LOPC	5	A		1.PSV-1001 set at 110barg (fire case)	

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	14.Less Temperature	1.See 1.2.1						
	15.Isolation, Depressuring or Blow-down	1.No issue						
	16.Start-up, shut down	1.No issue						
	17.Maintenance access							
	18.Containment, loss of							

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	19.Services, loss of	1.No issue						
	20.Noise, greater than threshold	1.Noise from BDV-B-1001	1.Hearing impairment	2	B			7.Confirm noise levels from BDV-CDT-001 and develop noise map as appropriate during DED
	21.General							
2.BBOU Open Drain & Closed Systems and Vent System	1.No Flow	1.Blockage on inlet to closed drain tank BBOU-CDT-001 (inadvertent closure of manual valves E2R	1.Loss of relief functionality when demand is placed on BDV-B-1001, and PSV-1001, potentially leading to rupture of the line and LOPC	5	B		1.SOP	9.Lock open E2R on the inlet to the BBOU-CDT-001
			2.Inability to drain the pig trap to the close drain system	2	B			8.Consider removing the valves E2R on vent line of BBOU-CDT-001

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
		2.Blockage on the vent line of BBOU-CDT-001 (inadvertent closure of manual valve E2R)	1.Loss of relief functionality, potentially leading to rupture of the line and LOPC	5	B		1.SOP	10.Lock open E2R on vent line of BBOU-CDT-001 8.Consider removing the valves E2R on vent line of BBOU-CDT-001
	2.More Flow	1.No issue						
	3.Less flow	1.No issues						
	4.Reverse Flow	1.N/A						
	5.As well as Flow	1.N/A						
	6.Part of Flow	1.N/A						
	7.Other than Flow	1.N/A						
	8.More Pressure	1.See flow Scenarios						

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	9.Less Pressure	1.See flow Scenarios						
	10.No Level	1.No issue						
	11.More Level	1.High level in BBOU-CDT-001	1.Potential liquid carryover to the vent system resulting into environmental pollution	3	C			11 Provide high level protection on BBOU-CDT-001
			2.Overpressure of BBOU-CDT-001 possibly leading to rupture	5	B			
			3.Loss of relief functionality on high level in the Tank	5	B			
		2.High liquid level on BBOU-ODT-001	1.Overfill of BBOU-ODT-001 leading to environmental pollution	3	C			12 Develop draining procedure for the pig trap

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	12.Less Level	1.No issue						
	13.More Temperature	1.No issue						
	14.Less Temperature	1.No issue						
	15.Isolation, Depressuring or Blow-down	1.No issue						
	16.Start-up, shut down	1.No issue						
	17.Maintenance access	1.See general Node						

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	18.Containment, loss of	1.See general Node						
	19.Services, loss of	1.No issue						
	20.Noise, greater than threshold	1.No issue						
	21.General							
3.ELEP MF Receivers & Launcher	1.No Flow	1.Blockage on inlet piping from SPDC ELEP wells (Fail close of SDV S1001/S1002: Inadvertent closure of any of the manual valves MV-1043, MV-1045,	1.Pressure build up upstream the blockage up the CITHP of SPDC ELEP well (383barg) potentially leading to rupture of the piping and LOPC, resulting into injuries/fatalities	5	C		1.Overpressure protection system from SPDC gathering manifold	1.Confirm the overpressure protection system on SPDC gathering manifold to ensure that is adequate to protect 900lb class piping
							2.PSV-1002 set at 110barg (fire case)	
							3.Pressure trip tied to ESD	
							4.PM	

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
		MV-1003, MV-1006)	from ignited release and environmental pollution from unignited release (Piping is 900lb class)				5.SOP	
		2.Blockage on BBOU pipeline into ELEPA manifold (Fail close of SDV S1002, Inadvertent closure of any of the manual valves MV-1059, MV-1017)	1.Pressure build up upstream the blockage up the CITHP of SPDC BBOU well (190barg) potentially leading to rupture of the piping and LOPC, resulting into injuries/fatalities from ignited release and environmental pollution from unignited release (Piping is 900lb class)	5	C		1.See 1.1.1	
		3.Blockage on BBOU/ELEPA	1.Pressure build up upstream	5	C		1.See 3.1.1	

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
		flow into ELEPA pig launcher (Inadvertent closure of any of the manual valves MV-1022, MV-1027, MV-1043, MV-1045, Fail close of SDV S1003)	the blockage up the CITHP of SPDC ELEPA well (383barg) potentially leading to rupture of the piping and LOPC, resulting into injuries/fatalities from ignited release and environmental pollution from unignited release (Piping is 900lb class)					
	2.More Flow	1.Fail open of BDV-B-1001	1.More gas to the close drain tank and vent system potentially leading to GHG emission	1	C		1.PM	2.Carry out ELEPA MF vent dispersion and thermal radiation modelling to determine safe dispersion distances
			2.Potential low temperature on the downstream piping (7 degC)	1	B		2.Piping is CS, LTD is -29degC	3.Install flame arrestor on the ELEPA MF vent tip

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
		2.Inadvertent opening of MV-1044 on the flow meter bypass	1.Metering error	1	B		1.SOP	
	3.Less flow	1.Passing relief valves (PSV-1002)	1.More gas to the vent system potentially leading to GHG emission	1	C		1.PM	
	4.Re-verse Flow	1.Possible re-verse flow from a future Nun River manifold into SPDC ELEPA when SPDC ELEPA is down	1.Potential process safety issues if there is maintenance work in SPDC ELEPA system	5	B			13Provide a check valve on all tie-in points into ELEPA manifold (class 1 check valve)
		2.Possible re-verse flow from BBOU pipeline into SPDC ELEPA when SPDC ELEPA is down or vice versa	1.Potential process safety issues if there is maintenance work in SPDC ELEPA or BBOU pipeline	5	B		1.See 3.4.1	
	5.As well as Flow	1.N/A						

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	6.Part of Flow	1.N/A						
	7.Other than Flow	1.N/A						
	8.More Pressure	1.See flow Scenarios						
	9.Less Pressure	1.See flow Scenarios						
	10.No Level	1.N/A						
	11.More Level	1.N/A						
	12.Less Level	1.N/A						
	13.More Temperature	1.Pool fire on the platform	1.Potential pressure build up on the piping possibly leading to rupture and LOPC	5	A		1.PSV-1002 set at 110barg (fire case)	

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	14.Less Temperature	1.See 3.2.1						
	15.Isolation, Depressuring or Blow-down							
	16.Start-up, shut down	1.No issue						
	17.Maintenance access	1.See general Node						
	18.Containment, loss of	1.See general Node						

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	19.Services, loss of	1.No issue						
	20.Noise, greater than threshold	1.No issue						
	21.General							
4.ELEP Open Drain, Closed Drain System and Vent System	1.No Flow	1.Blockage on inlet to closed drain tank ELEP-CDT-001 (inadvertent closure of manual valves E2R	1.Loss of relief functionality when demand is placed on BDV-B-1001, and PSV-1001, potentially leading to rupture of the line and LOPC	5	B		1.SOP	16.Lock open E2R on the inlet to the ELEP-CDT-001
			2.Inability to drain the pig trap to the close drain system	5	B			8.Consider removing the valves E2R on vent line of BBOU-CDT-001

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
		2.Blockage on the vent line of ELEP-CDT-001 (inadvertent closure of manual valve E2R)	1.Loss of relief functionality, potentially leading to rupture of the line and LOPC	5	B		1.SOP	17 Lock open E2R on vent line of ELEP-CDT-001 8.Consider removing the valves E2R on vent line of BBOU-CDT-001
	2.More Flow	1.No issue						
	3.Less flow	1.No issue						
	4.Reverse Flow	1.N/A						
	5.As well as Flow	1.N/A						
	6.Part of Flow	1.N/A						
	7.Other than Flow	1.N/A						
	8.More Pressure	1.See flow Scenarios						

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	9. Less Pressure	1.No issue						
	10.No Level	1.N/A						
	11.More Level	1.High level in ELEP-CDT-001	1.Potential liquid carryover to the vent system resulting into environmental pollution	3	C			18Provide high level protection on ELEP-CDT-001
			2.Overpressure of ELEP-CDT-001 possibly leading to rupture	5	B			
			3.Loss of relief functionality on high level in the Tank	5	B			
		2.High liquid level on ELEP-ODT-001	1.Overfill of ELEP-ODT-001 leading to environmental pollution	3	B			12Develop draining procedure for the pig trap

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	12.Less Level	1.N/A						
	13.More Temperature	1.No issue						
	14.Less Temperature	1.No issue						
	15.Isolation, Depressuring or Blow-down	1.See general Node						
	16.Start-up, shut down	1.No issue						
	17.Maintenance access	1.See general Node						

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	18.Containment, loss of	1.See general Node						
	19.Services, loss of	1.No issue						
	20.Noise, greater than threshold	1.N/A						
	21.General							
5.AKABEL Pig Receiver	1.No Flow	1.Blockage on line to AKABEL slug catcher (fail close of SDV-S1001, inadvertent closure of manual valve M-1043, MV-1045, MV-1059)	1.Pressure build up upstream the blockage up the CITHP of SPDC ELEPA well (383barg) potentially leading to rupture of the piping and LOPC, resulting into injuries/fatali-	5	C		1.See 3.1.3	

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
			ties from ignited release and environmental pollution from unignited release (Piping is 900lb class)					
	2. More Flow	1. Inadvertent opening of MV-1044 on the flowmeter bypass	1. Metering error	1	C		1. SOP	
	3. Less flow	1. No issue						
	4. Reverse Flow	1. N/A						
	5. As well as Flow	1. N/A						
	6. Part of Flow	1. N/A						
	7. Other than Flow	1. N/A						

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	8.More Pressure	1.See flow Scenarios						
	9.Less Pressure	1.See flow Scenarios						
	10.No Level	1.N/A						
	11.More Level	1.N/A						
	12.Less Level	1.N/A						
	13.More Temperature	1.N/A						
	14.Less Temperature	1.N/A						
	15.Isolation, Depressuring or	1.See general Node						

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	Blow-down							
	16.Start-up, shut down	1.See general Node						
	17.Maintenance access	1.See general Node						
	18.Containment, loss of	1.See general Node						
	19.Services, loss of	1.N/A						
	20.Noise, greater than threshold	1.N/A						
	21.General							

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
6.GENERAL	1.No Flow	1.No adequate maintenance access for safety critical equipment	1.Potential loss of integrity for safety critical equipment leading to increased probability of failure on demand	5	B			6.Carry out 3D model review for the entire project facility during DED
	2.More Flow							
	3.Less flow							
	4.Re-verse Flow							
	5.As well as Flow							
	6.Part of Flow							
	7.Other than Flow							
	8.More Pressure							

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	9.Less Pressure							
	10.No Level							
	11.More Level							
	12.Less Level							
	13.More Temperature							
	14.Less Temperature							
	15.Isolation, Depressuring or Blow-down	1.No means of positive isolation	1.Potential process safety risk during maintenance	5	B			14 Provide the means for positive isolation of the incoming lines into ELEPA manifold
		2.No means of manual depressurization	1.same as 3.2.1					15 Provide a means of manual depressurization of the ELEPA header

Node	Deviation	Cause	Consequence	Before Risk Reduction			Effective Safeguards	Recommendations
				S	L	RR		
	16.Start-up, shut down							
	17.Maintenance access							
	18.Containment, loss of	1.Loss of primary containments (leaks from flanges and instrument fittings, corrosion leaks, etc.)	1.Release of gas to the environment potentially leading to fire and explosion from ignited release and environmental pollution from unignited release	5	C		1.FLM	21 Carry out hazardous area classification of the entire project facilities
							2.Layout (Separation distances)	22 Develop Emergency Spill Response Procedure/Plan for the project facilities
							3.Gas detection tied to ESD	23 Provide fire detection system in all the manifold
	19.Services, loss of							

[illegible]











